YAKEEN

NEET 2.0 2027

Physics | Lecture - 6

Motion in a Plane

By- Manish Raj (MR Sir)



Todays Goal

- Projectile on Inclined. ✓
- Relative motion in 1-D ✓

H/w goal

- ✓ Maha-Manthan - 1-D
- ✓ S.A (2-D) → 1, 2
- ✓ → Questin in (Notes) this

Question

Find the value of 'u' so that the ball reaches at point B. (Take $g = 10\ m/s^2$)

1. 20 m/s
2. 40 m/s
3. $15\sqrt{2}$ m/s
4. 50 m/s

Consider motn in x-axis

$15 = u_x \cdot tf$

motn in y-axis

$10 = u_y t - \frac{1}{2} g t^2$

Put $tf = \frac{15}{u_x}$

$10 = u_y \frac{15}{u_x} - \frac{1}{2} g \left(\frac{15}{u_x}\right)^2$

$10 = \tan\theta \times 15 - \frac{10}{2} \frac{15 \times 15}{u_x^2}$

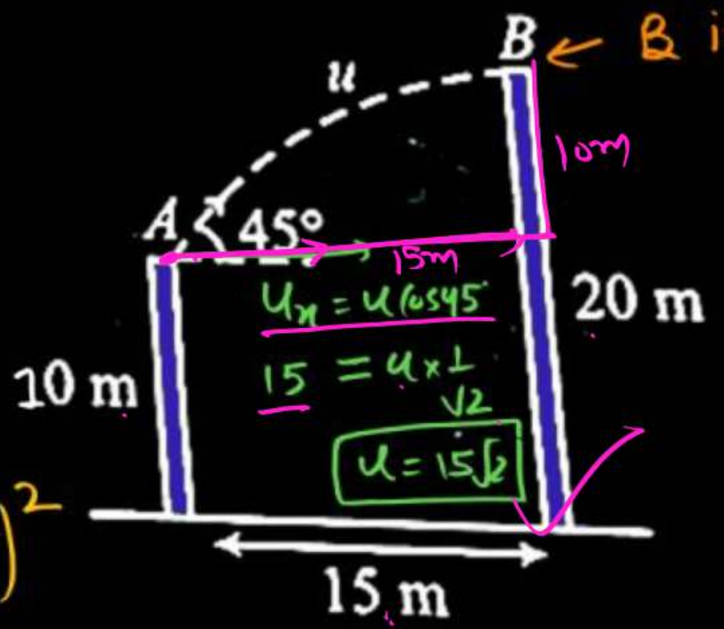
$10 = 15 - \frac{5(15)^2}{u_x^2}$

$\frac{5(15)^2}{u_x^2} = 15 - 10$

$\frac{5(15)^2}{u_x^2} = 5$

$u_x^2 = 15^2$

$u_x = 15$

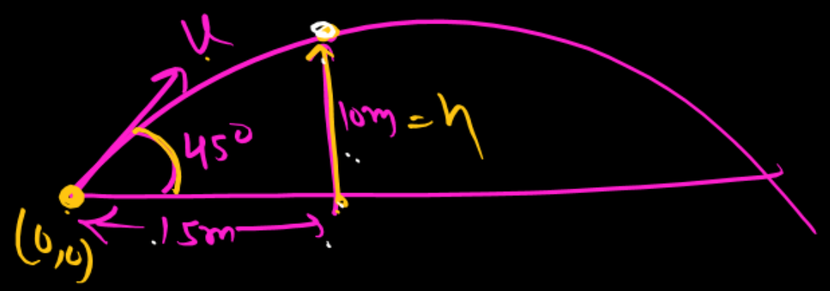


$u_x = u\cos 45°$

$15 = u \times \frac{1}{\sqrt{2}}$

$u = 15\sqrt{2}$

2nd method



$$y = x\tan\theta\left(1-\frac{x}{R}\right)$$

$$10 = 15\tan 45°\left(1-\frac{15}{R}\right) \checkmark$$

$$\frac{2}{3} = 1-\frac{15}{R}$$

$$\frac{15}{R} = 1-\frac{2}{3} = \frac{1}{3}$$

$$R = 45m = \frac{u^2\sin 2\theta}{g}$$

$$45 = \frac{u^2\sin(2\times 45)}{10}$$

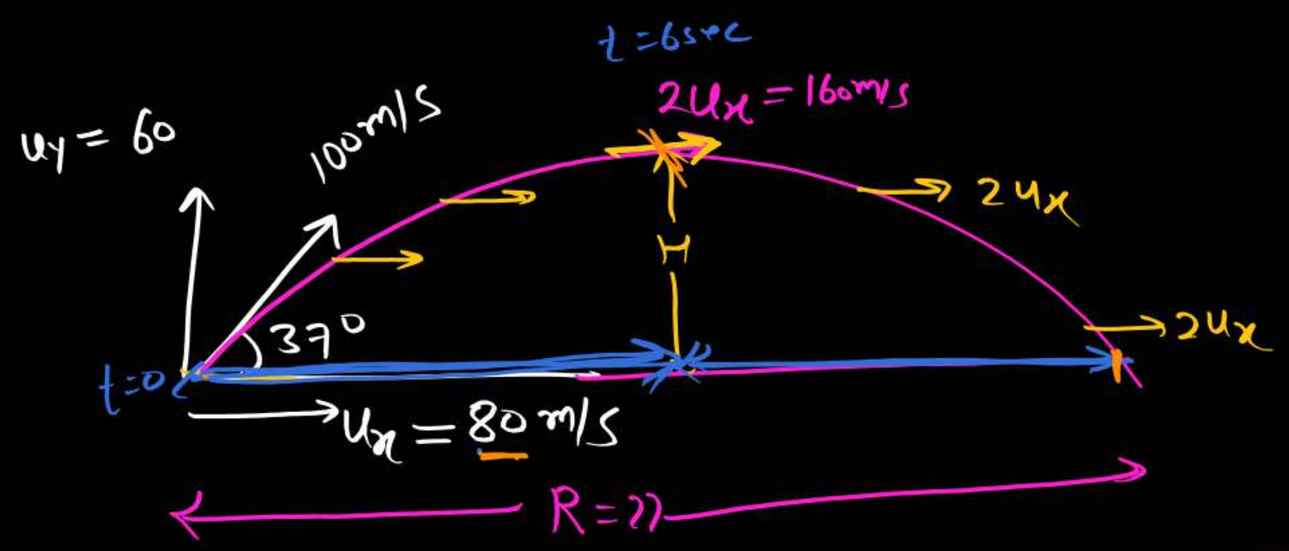
$$450 = u^2 \qquad u = \sqrt{450}\ m/s$$

$$= \sqrt{225\times 2} = 15\sqrt{2}\ m/s$$

Copy

⑥ Ball is projected with 100m/s at angle 37° at maximum height Impulse imparted to the Ball so that its velocity becomes double as it's maximum height then find T_f & total Range

JEE-Adv



$$R' = x_1 + x_2$$
$$= 80\times 6 + 160\times 6$$
$$= 480\text{m} + 960\text{m}$$
$$R' = 1440\text{ m}$$

$H_{max} = \frac{u_y^2}{2g} = 180\text{m}$

$T_f = \frac{2u_y}{g} = \frac{2\times 60}{10} = 12\text{sec}$ (same)

$R(\text{initial}) = 960\text{m}$

Question

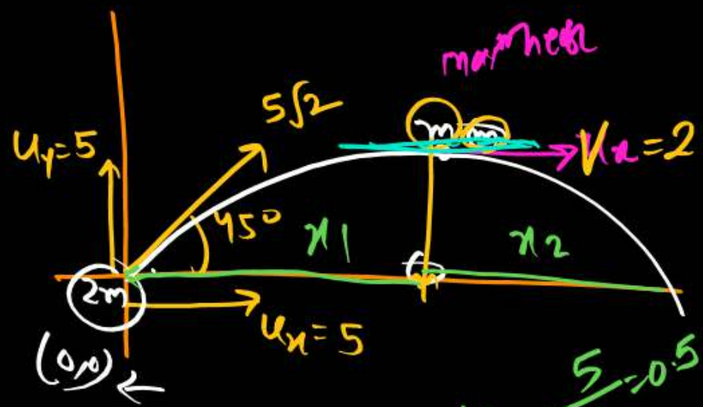
Read & Try
No Copy

A projectile is thrown from a point O on the ground at an angle 45° from the vertical and with a speed $5\sqrt{2}$ m/s. The projectile at the highest point of its trajectory splits into two equal parts. One part falls vertically down to the ground 0.5 s after the splitting. The other part, t seconds after the splitting, falls to the ground at a distance x meters from the point O. The acceleration due to gravity $g = 10$ m/s^2.

[JEE Adv. 2021]

The value of t is 0.5sec.

The value of x is 7.5m.



$T_{up} = \frac{dy}{g} = \frac{5}{10} = 0.5$

$x = x_1 + x_2$
$= 5\times0.5 + 10\times0.5$
$= 2.5 + 5 = 7.5$m

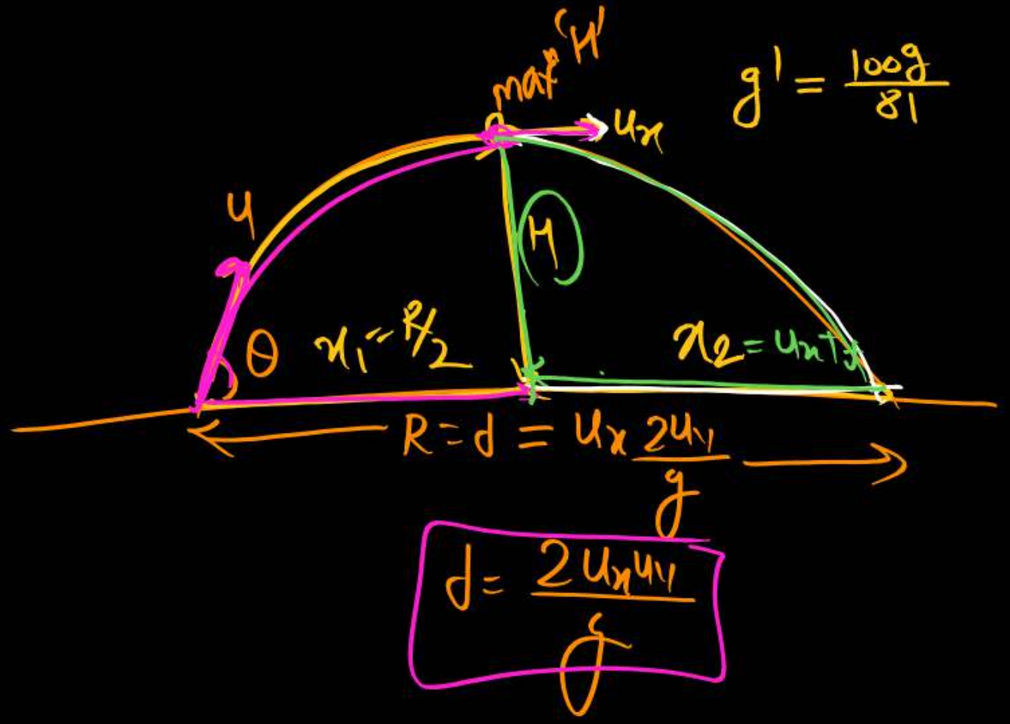
UEP

$P_i x = P_f x$
$2m\,u_x = m\times0 + m\,v$
$v = 2u_x$ — (1)

Question

copy at ghar

A projectile is fired from horizontal ground with speed v and projection angle θ. When the acceleration due to gravity is g, the range of the projectile is d. If at the highest point in its trajectory, the projectile enters a different region where the effective acceleration due to gravity is $g' = \frac{g}{0.81}$ then the new range is $d' = nd$. The value of n is _____. [JEE Adv. 2022]



$g' = \frac{100g}{81}$

$d = \frac{2u_x u_y}{g}$

from Height to ground $T_f = \sqrt{\frac{2H}{g'}} = \sqrt{\frac{2u_y^2}{\frac{100g}{81}}} = \frac{9u_y}{10g}$

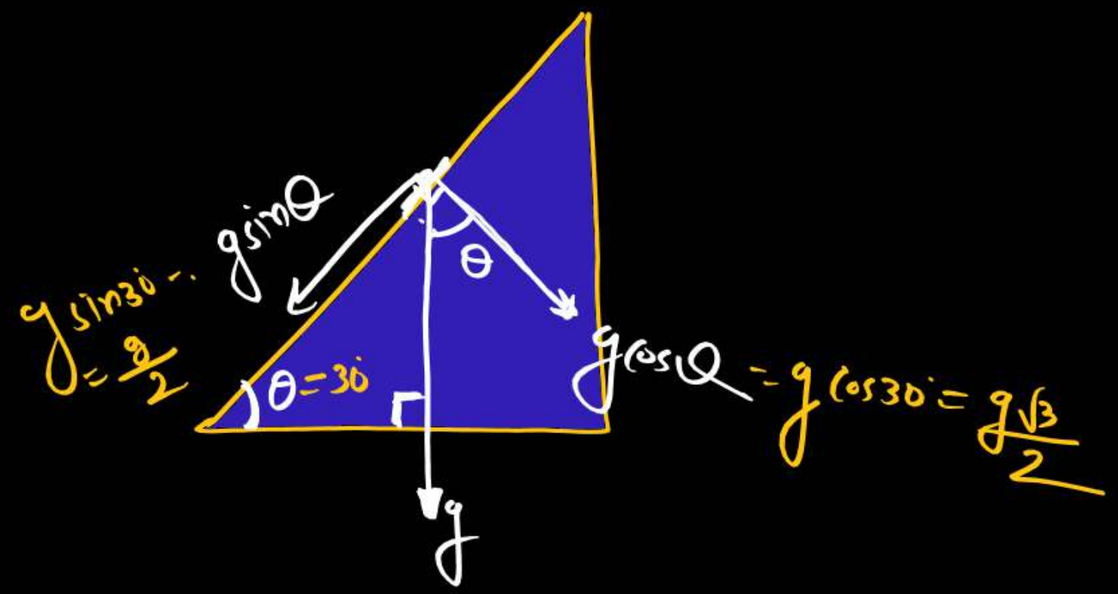
$x_2 = u_x \times T_f = \frac{u_x 9u_y}{10g} = \frac{9u_x u_y}{10g}$

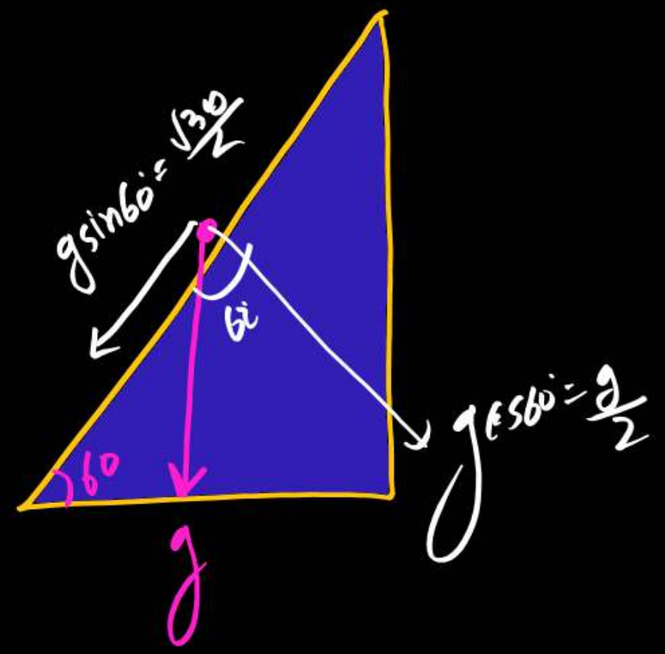
$n = \frac{d'}{d} = \frac{\frac{19u_x u_y}{10g}}{\frac{2u_x u_y}{g}}$

$n = \frac{19}{20}$ JJE Adv

$d' = \frac{R}{2} + \frac{9u_x u_y}{10g}$

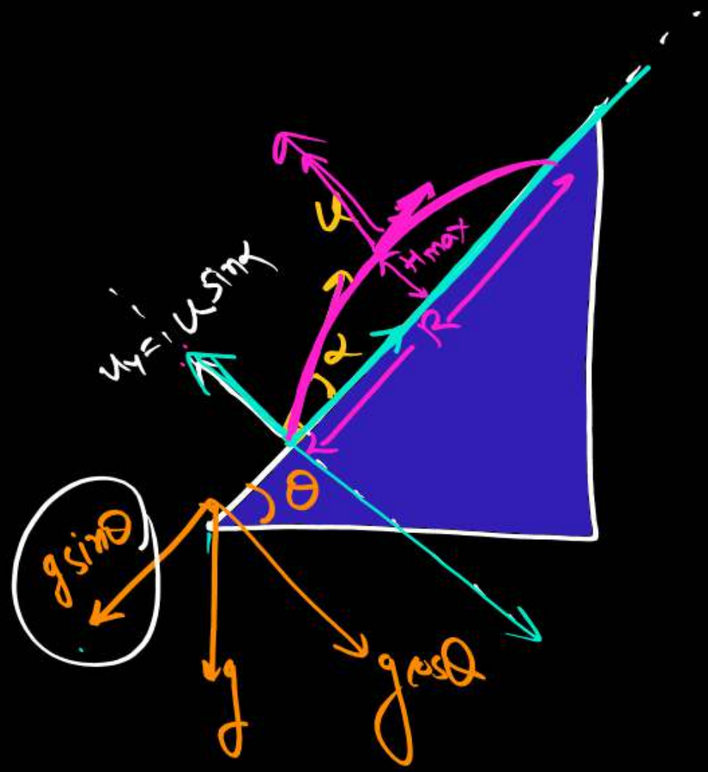
$= \frac{u_x u_y}{g} + \frac{9}{10}\frac{u_x u_y}{g} = \frac{19u_x u_y}{10g}$

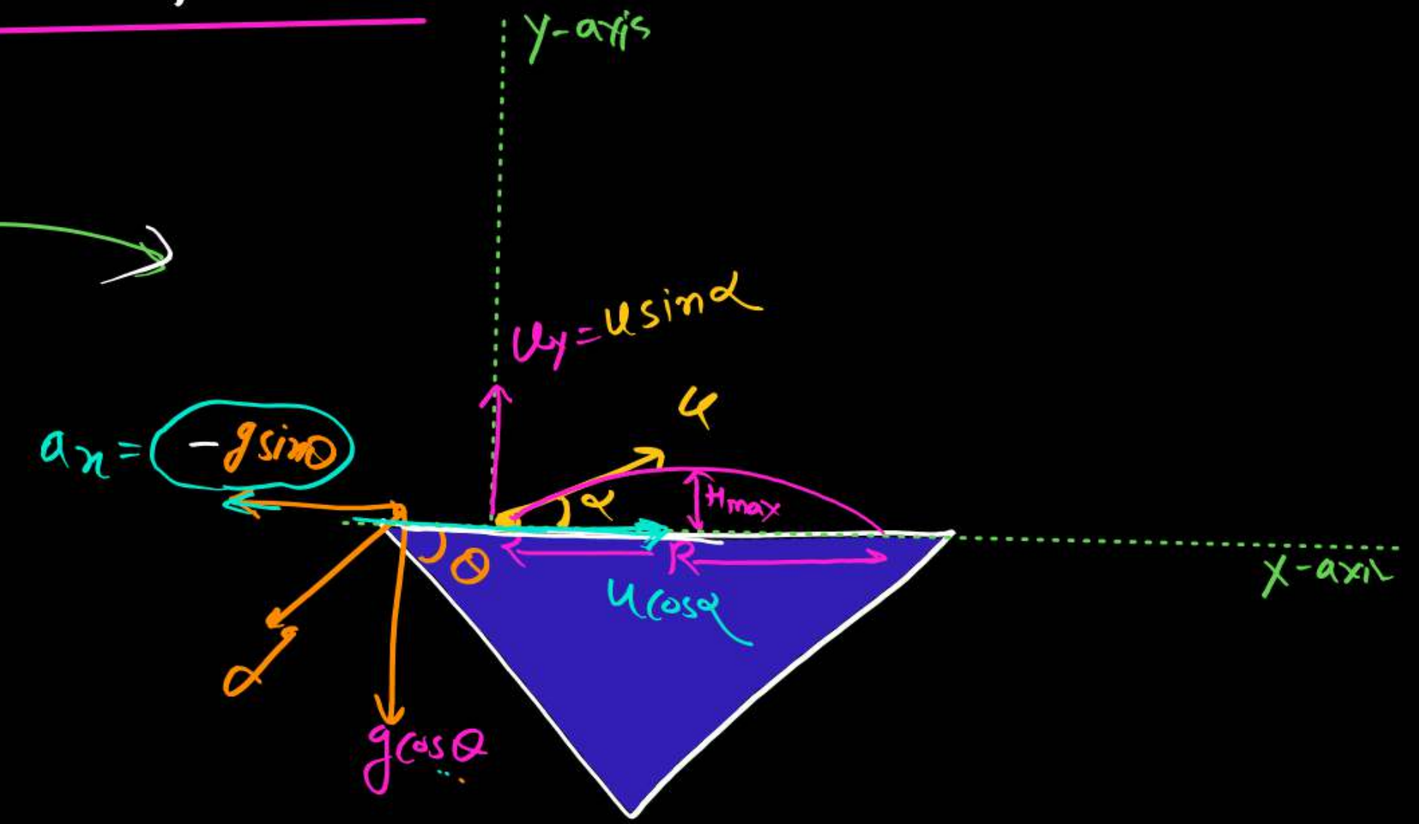

$g\sin\theta$
$g\sin30^\circ = \frac{g}{2}$
θ
$\theta = 30^\circ$
$g\cos\theta = g\cos30^\circ = \frac{g\sqrt{3}}{2}$
g


$g\sin60^\circ = \frac{\sqrt{3}g}{2}$
60°
60°
$g\cos60^\circ = \frac{g}{2}$
g

Projectile motion on an Inclined plane :-

T_f, R, H_{max}





Motⁿ in vertical dirⁿ (like Motⁿ under gravity)

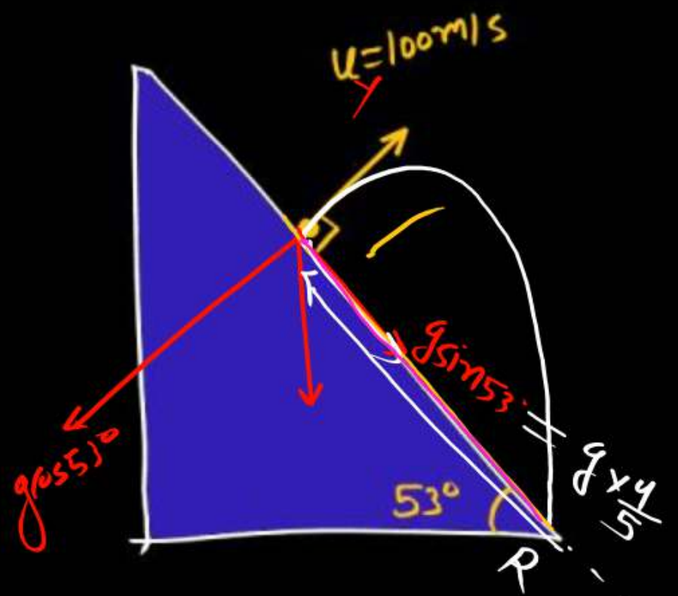
$$T_f = \frac{2u_y}{g\cos\theta} = \frac{2u\sin\alpha}{g\cos\theta}$$

$$H_{max} = \frac{u^2\sin^2\alpha}{2g\cos\theta}$$

$$\text{Range} = u\cos\alpha\, T_f - \frac{1}{2}g\sin\theta\,(T_f)^2$$

Ball is Projected ⊥r to Inclined plane then find T_f and Range ??

$$T_f = \frac{2u_y}{g_1} = \frac{2\times 100}{10\times\frac{3}{5}} = \frac{100}{3}\text{sec}$$



$g_x = g\times\frac{4}{5}$

$u_x = 0$

$$\text{Range} = u_x T_f + \frac{1}{2}\left(\frac{4g}{5}\right)T_f^2$$

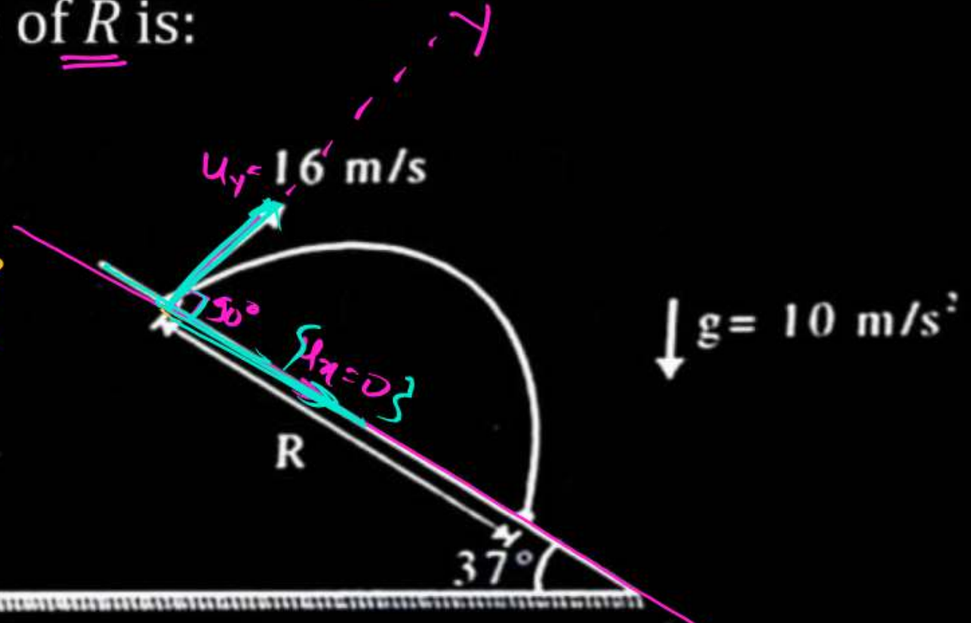
$$= \left(\frac{1}{2}\times\frac{4\times10^2}{5}\times\frac{100\times100}{9}\right)m$$

$$= \frac{4\times10^4}{9}m$$

Question

A projectile is launched with a velocity of 16 m/s at right angles to the slope which is inclined at 37° with the horizontal. The value of R is:

1. 96 m
2. {48 m} Ans
3. 72 m
4. None of these



$$T_f = \frac{2u_y}{g\cos 37^\circ} = \frac{2 \times 16}{10 \times \frac{4}{5} \times \frac{1}{2}}$$

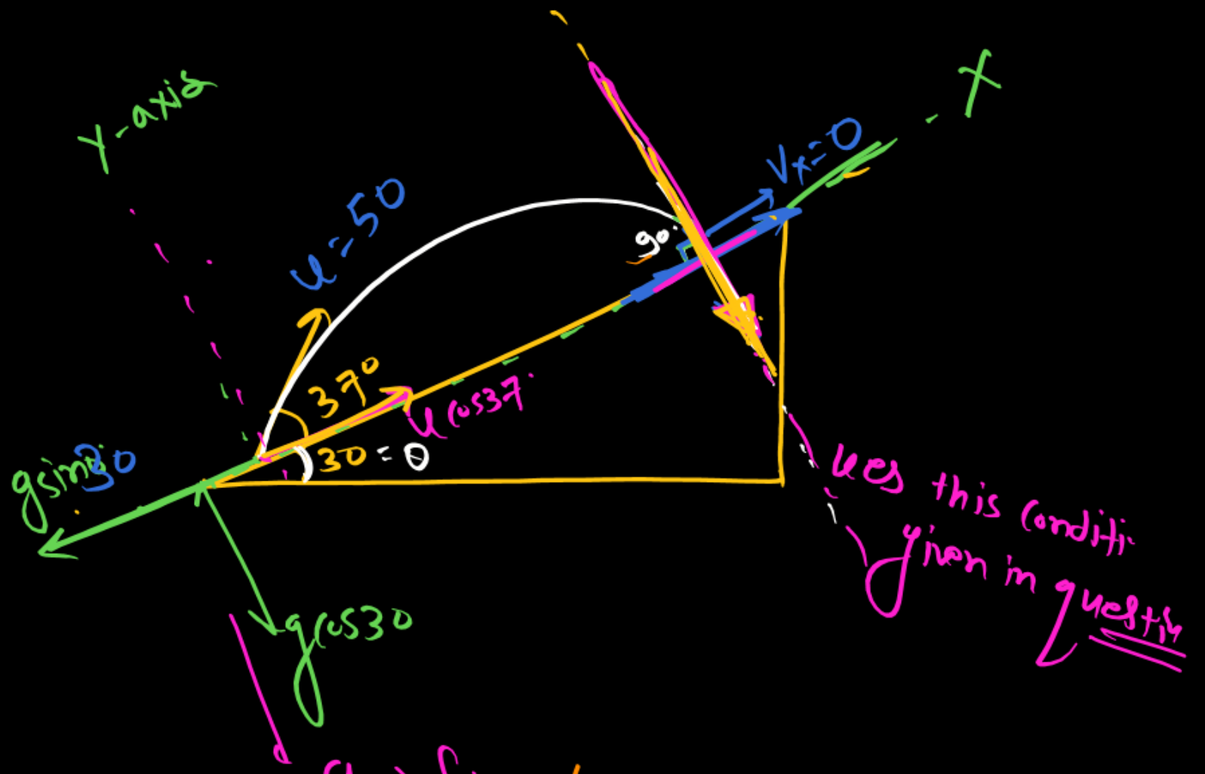
$$T_f = 4 \text{ sec}$$

$$R_{aye} = \frac{1}{2} g \times \frac{3}{5} (4)^2$$

$$= \frac{1}{2} \times 10 \times \frac{3}{5} \times 16$$

$$= 48 \text{ m}$$

If Ball collide perpendicular to Inclined plane then find T_f = ??



(due to extra air friction T_f ka formula Nahi layega)

motion in X-axis only

$$\vec{V}_x = \vec{u}_x + a_x t$$

$$0 = 50\cos 37° - g\sin 30\, t$$

$$10 \times \frac{1}{2} t = \frac{10}{50} \times \frac{4}{5}$$

$$t = \frac{40}{5} = 8\ sec$$

H/w find relation B/w α, β and θ.

ghar pe rough

$$\tan\theta = \tan\alpha + \tan\beta$$

done

hint eqn of Trajectory

$$H = x_1 \tan\theta\left(1 - \frac{x_1}{x_1+x_2}\right)$$

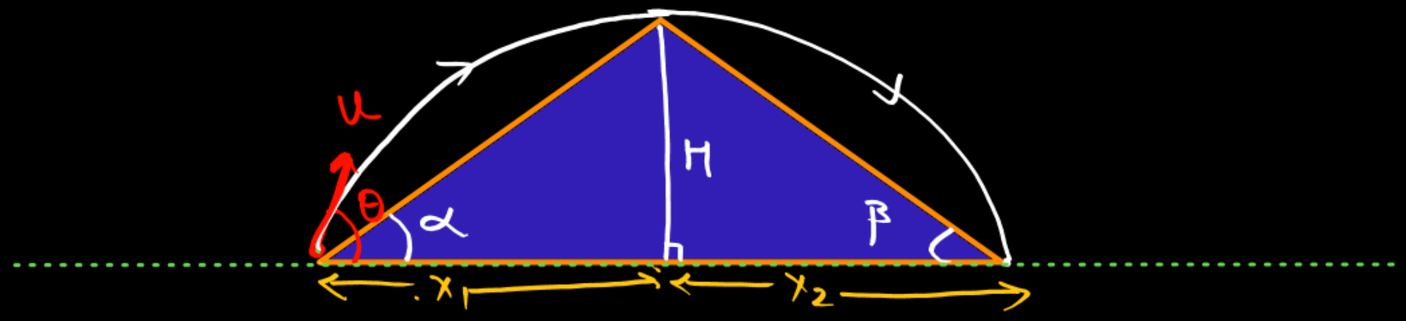
$$H = x_1 \tan\theta\left(\frac{x_1+x_2-x_1}{x_1+x_2}\right)$$

$$H = x_1 \tan\theta\left(\frac{x_2}{x_1+x_2}\right)$$

$$H = \frac{x_1 x_2 \tan\theta}{x_1+x_2}$$

$$H\left(\frac{x_1+x_2}{x_1 x_2}\right) = \tan\theta$$

$$H\left(\frac{1}{x_1}+\frac{1}{x_2}\right) = \tan\theta$$



$$\tan\alpha = \frac{H}{x_1} \qquad \tan\beta = \frac{H}{x_2}$$

$$y = x\tan\theta\left(1 - \frac{x}{R}\right)$$

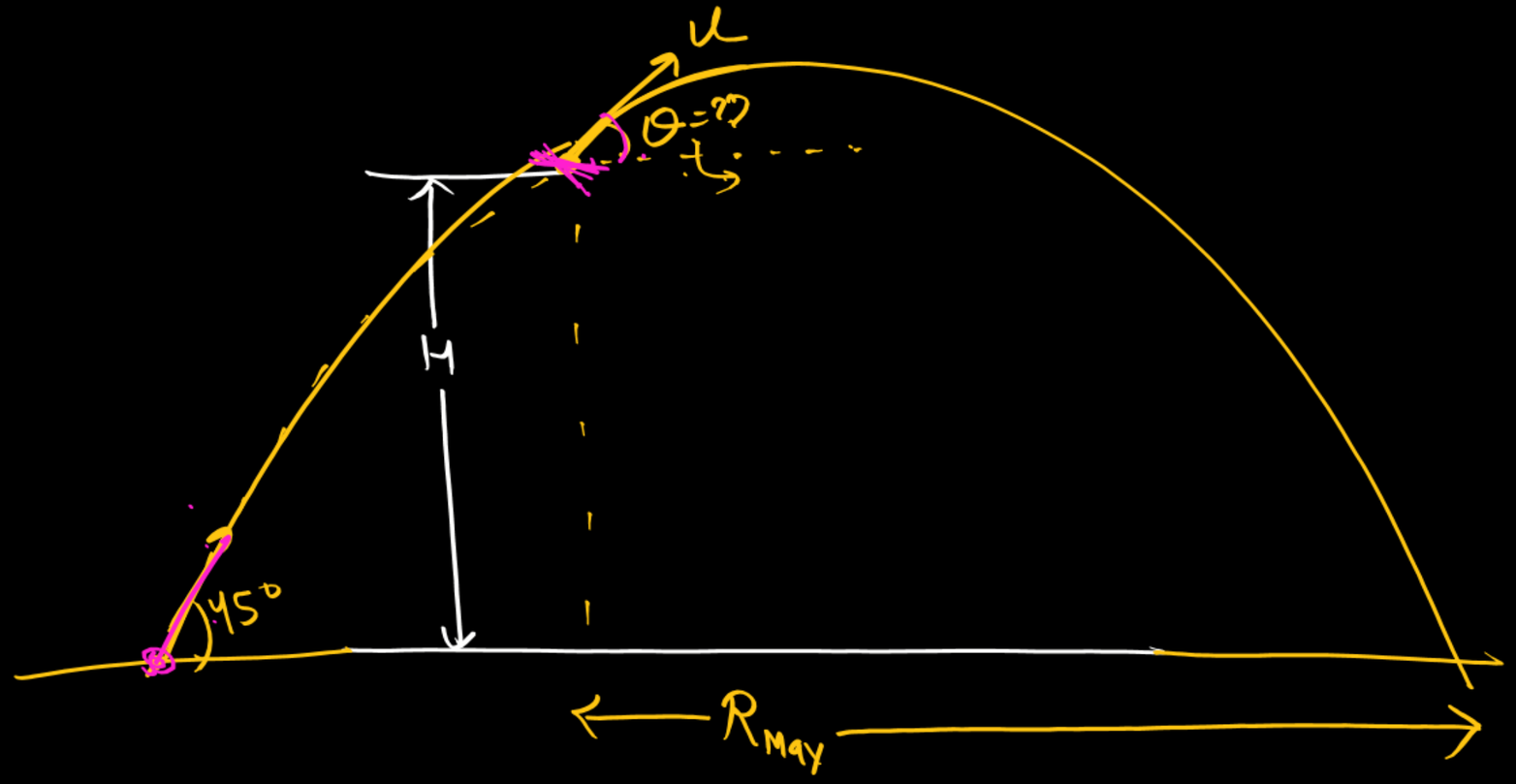
$$\tan\alpha + \tan\beta = \tan\theta$$

Ball is Projected from Tower then find Angle at which it is Projected to have maximum Horizontal Range.

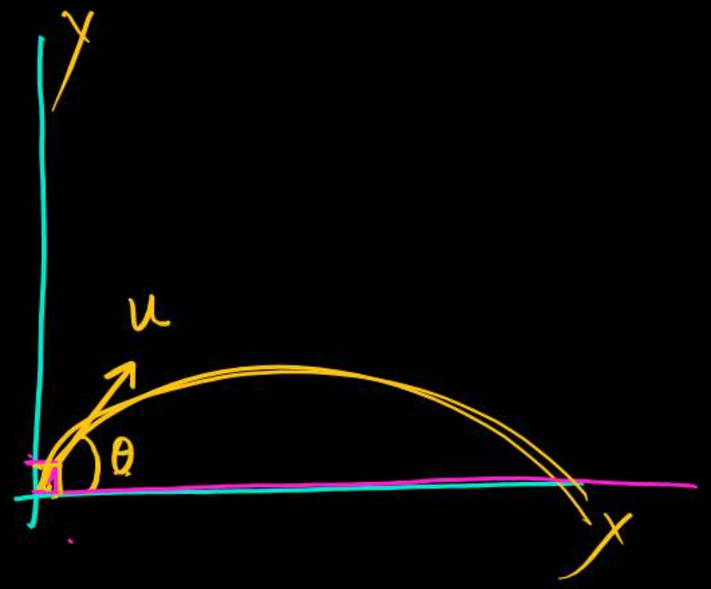
(a) $\theta = 45$

(b) $\theta > 45^\circ$

(c) $\theta < 45^\circ$



Case-1
Ground to Ground

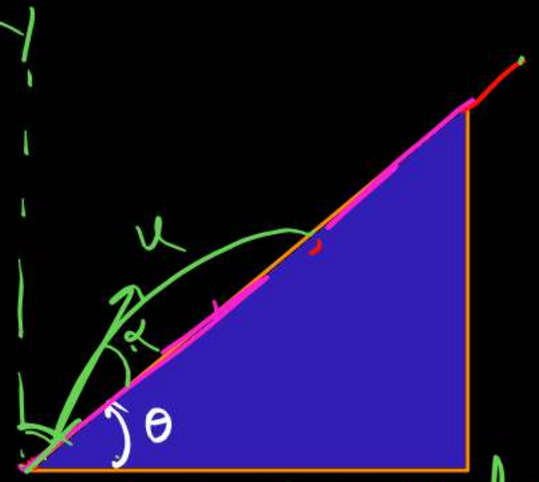


$\theta = \frac{90}{2} = 45^\circ$ for

R_{max}

Not for

Case-2
Up to Inclined



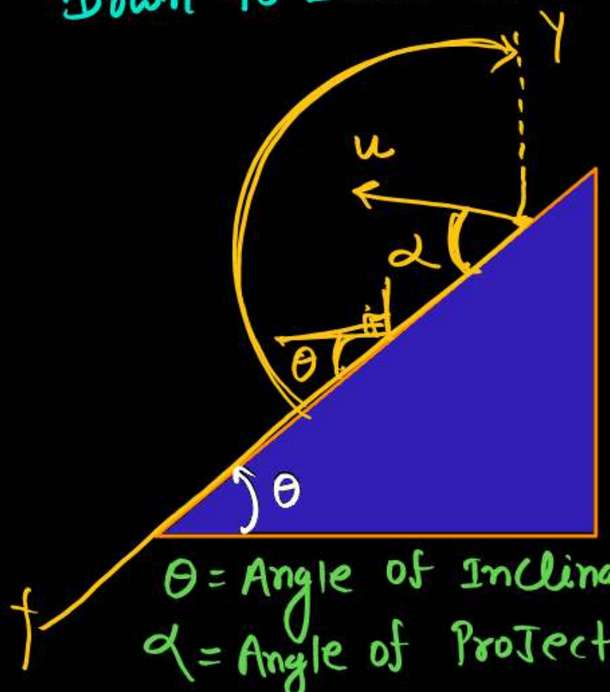
θ = Angle of Inclination

α = Angle of Projection

find α for maximum Range.

$$\alpha = \left[\frac{90-\theta}{2}\right]$$

Case-3
Down to Inclined



θ = Angle of Inclination

α = Angle of Projection

$$\alpha = \left[\frac{90+\theta}{2}\right]$$

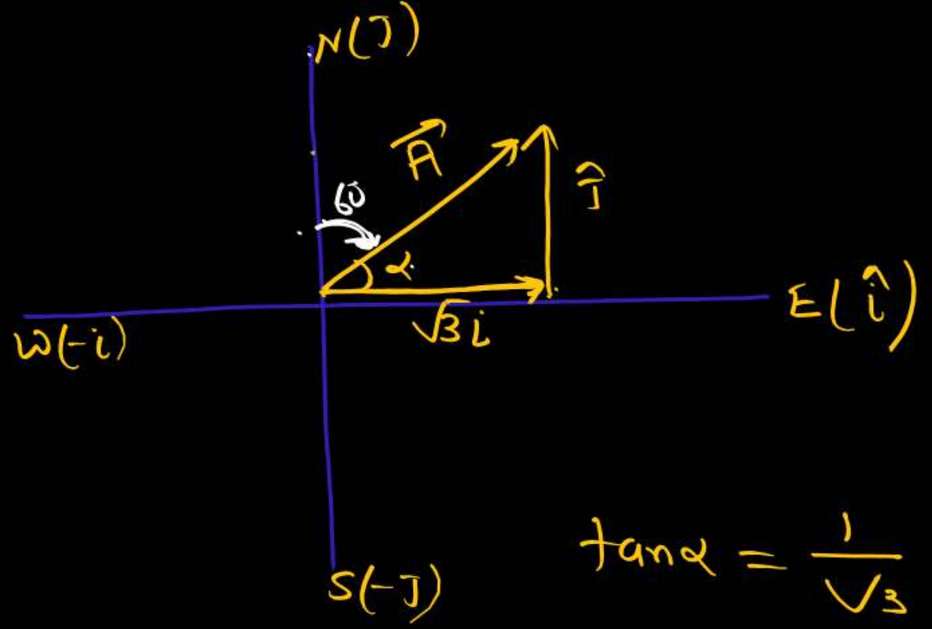
How To draw vector

MR* Box → Jab bhi vector ka direction likha ho, X-Y Plane me vector ko draw karo then use

$$\left[\tan\theta = \frac{P}{B}\right];$$

Find dirⁿ of vector from x-y axis.

① $\vec{A} = \sqrt{3}\,\hat{i} + \hat{j}$

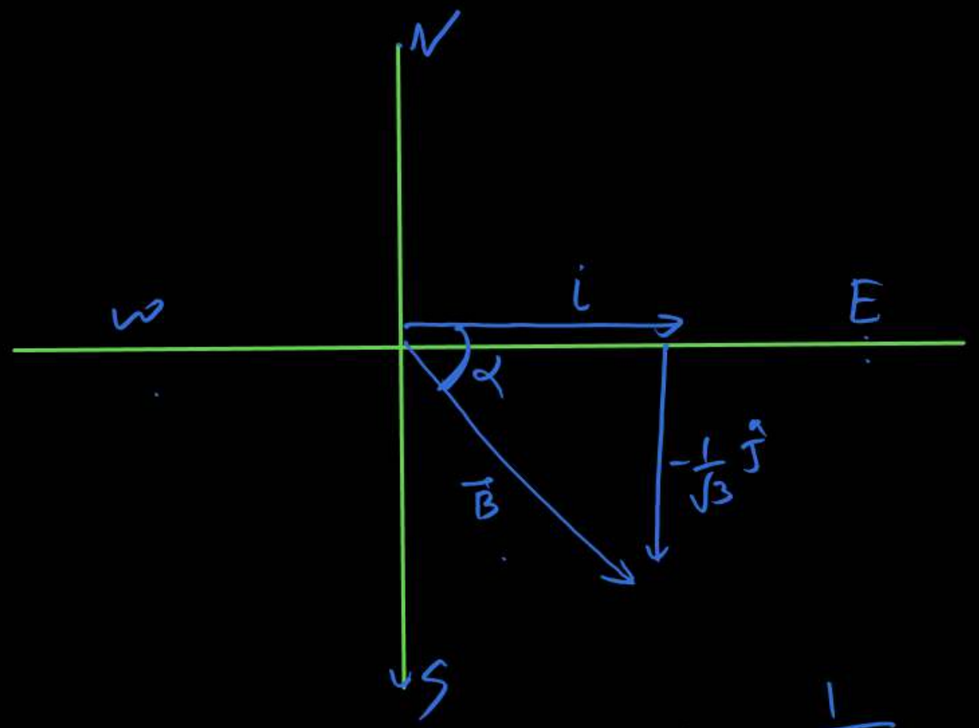


$\tan\alpha = \frac{1}{\sqrt{3}}$

$\alpha = 30°$

$\vec{A}$ is 30° North of east
A is 60° east of North

② $\vec{B} = i - \frac{1}{\sqrt{3}}\hat{j}$



$\tan\alpha = \frac{1}{\sqrt{3}\times 1}$

$\alpha = 30$

$\vec{B}$ is 30° South of east
$\vec{B}$ is 60° east of south

find dirn of vector from x-y axis.

H/W

③ $\vec{c} = -i - \sqrt{3}\,\hat{j}$

④ $\vec{D} = 2i - \sqrt{3}\,\hat{j}$

find dirⁿ of vector from x-y axis.

⑤ $\vec{E} = -3\hat{i} + \sqrt{3}\hat{j}$

⑥ $\vec{F} = +3i - \sqrt{3}\hat{j}$

Special case in Relative motion in 1-D

(1) Train ↔ man/Bird Prop^{m}

(2) Train Prob^{m} [crossing/overtaking]

(3) Bus - CAR overtaking

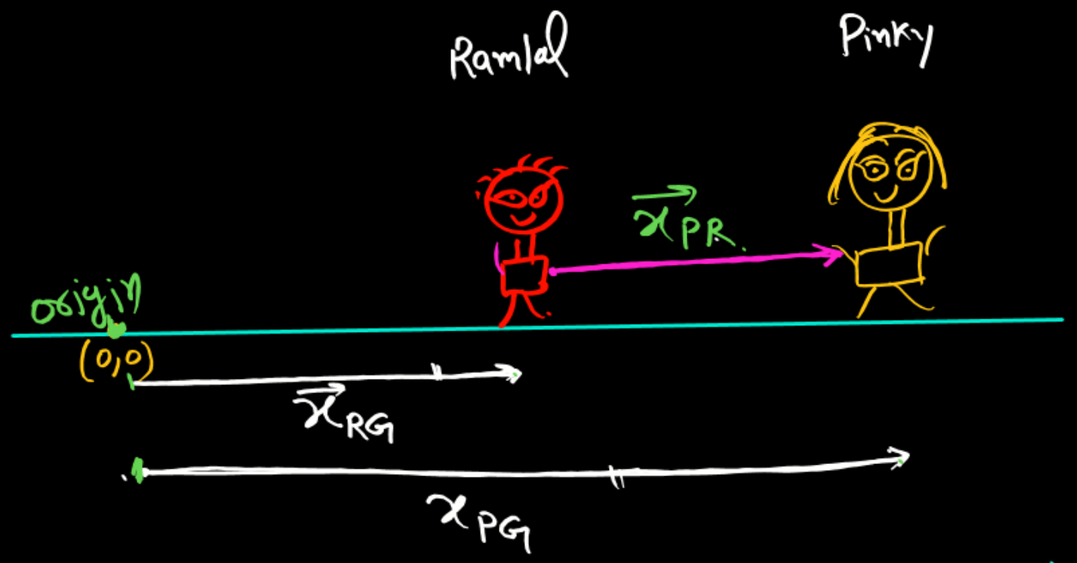
(4) Escalator Prop^{m}

(5) Upstream & downstream in river

(6) Realive motion under gravity

X

Relative Motion In 1-D



$$\vec{x}_{PR} = \vec{x}_{PG} - \vec{x}_{RG} = \vec{x}_{PG} + (-\vec{x}_{RG})$$

Position of Pinky w.r.t. Ramlal

diff^n w.r.t time

$$\vec{V}_{PR} = \vec{V}_{PG} - \vec{V}_{RG}$$

$\vec{V}_{PR}$: Velocity of Pinky w.r.t Ram

$\vec{V}_{PG}$: velocity of Pinky w.r.t ground

$\vec{V}_{RG}$: velocity of Ramly

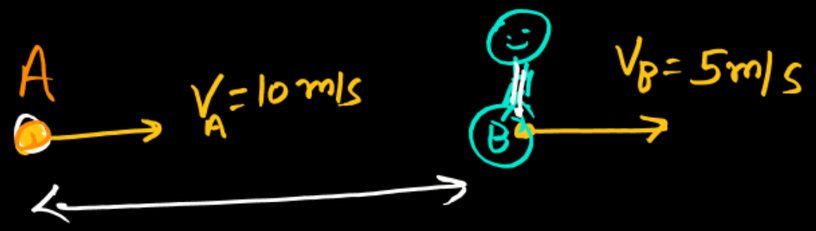
$$\vec{V}_{AB} = \vec{V}_{AG} - \vec{V}_{BG}$$

$$\boxed{\vec{V}_{RP} = \vec{V}_{RG} - \vec{V}_{PG}}$$

diff^n w.r.t. time

$$\vec{a}_{PR} = \vec{a}_{PG} - \vec{a}_{RG}$$

$\vec{a}_{PR}$: acc^n of Pinky w.r.t Ramlal

① 

$$\vec{V}_{AB} = V_{Ag} - V_{Bg}$$

$$= 10i - 5i$$

$$\vec{V}_{AB} = 5\hat{i}$$

↑ Observer

$$\vec{V}_{BA} = \vec{V}_B - \vec{V}_A$$

$$= 5i - 10i$$

$$= -5i\ m/s$$

←

$V_{AA} = 0$

$V_{BB} = 0$

Observer khud ko rest me manta hai

② 

$$\vec{V}_{AB} = \vec{V}_A - \vec{V}_B$$

$$= 10 - 20 = -10\hat{i}$$

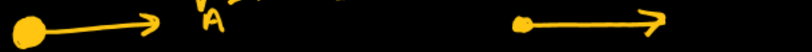
↓

Velocity of A w.r.t B

$$\vec{V}_{BA} = \vec{V}_B - \vec{V}_A$$

$$= 20 - 10$$

$$= 10\hat{i}$$

① 

$$v_{AB} = 0$$
$$v_{BA} = 0$$



$$\vec{v}_{AB} = \vec{V}_A - \vec{V}_B$$

$$= 10 - (-20)$$

$$\vec{V}_{AB} = 30\hat{i}$$

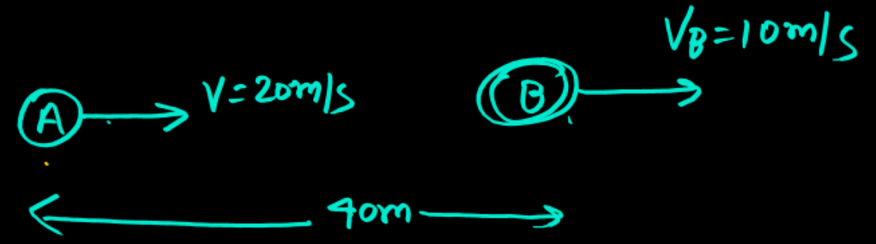
$$\vec{V}_{BA} = -30\hat{i}$$

$$\vec{v}_{AB} = -\vec{V}_{BA}$$

MR* BOX:

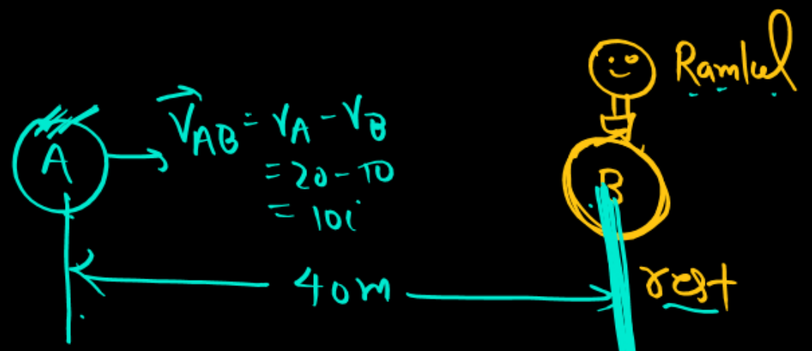
→ observer khud ko rest me hi assume karta hai

→ Jis object ka relative velocity likhna hai, ush object ka velocity as it is lo and observer ki velocity ulta(-ve) le ke usme chipka (add) kar do.



find time when they meet ??

Soln

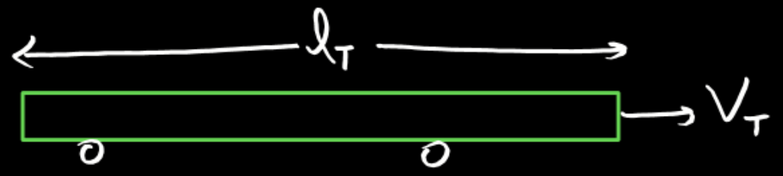


$$t = \frac{40}{10} = 4 \text{ sec}$$

(c)" at a·

MR* Box

Ramlal obseber ban kar Jis object Pe baith Jayga usko rest me manega, & object ki velocity ulta kar ke add kar dega Jisko (other object) dekhega.





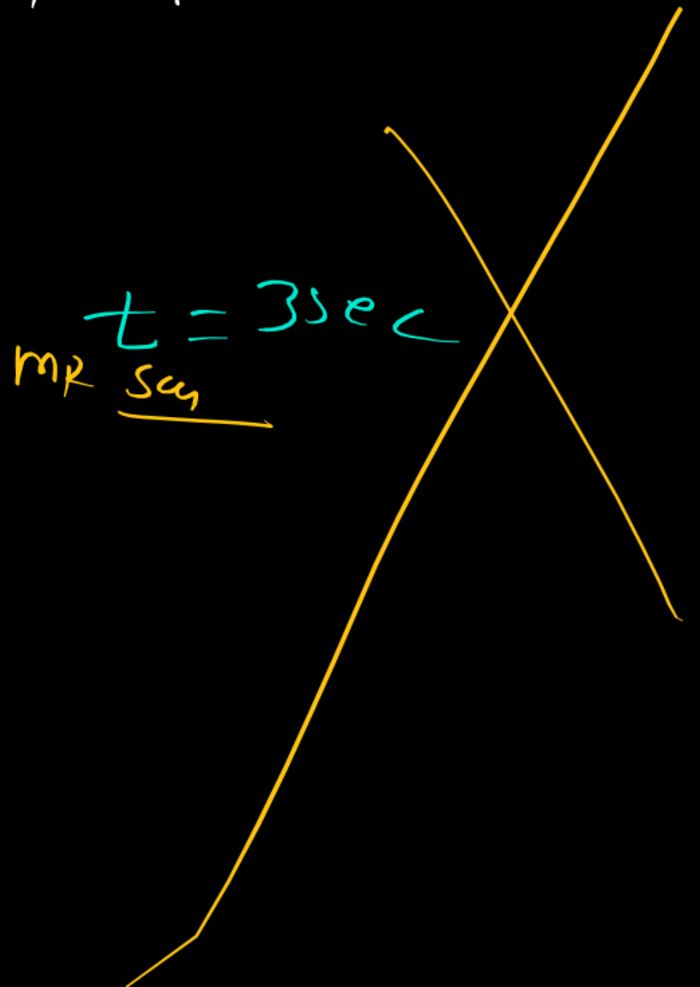
Find time taken by train to cross man.

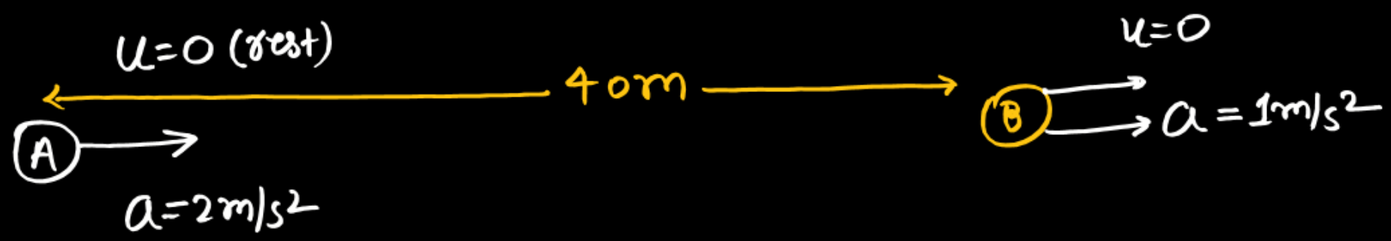


time when they meet

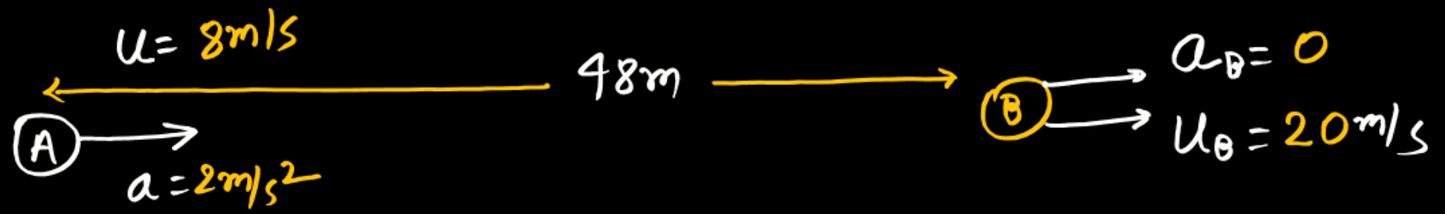
$t = 3\,sec$

mR Sam




u=0 (rest)
40m
A
a=2m/s²
u=0
B
a=1m/s²


u=60m/s
80m
A
a=0
B
$a_B = 10 m/s^2$
$u_B = 0$


u = 8 m/s
48 m
A
a = 2 m/s²
B
a_B = 0
u_B = 20 m/s

Question

Two trains each of length 100 m moving parallel towards each other at speed 72 km/h and 36 km/h respectively. In how much time will they cross each other?

1. 4.5 s
2. 6.67 s
3. 3.5 s
4. 7.25 s

Question

A bus is moving with a speed of 10 ms^{-1} on a straight road. A scooterist wishes to overtake the bus in 100 s. If the bus is at a distance of 1 km from the scooterist, with what speed should the scooterist chase the bus? **[2009]**

1. 40 ms^{-1}
2. 25 ms^{-1}
3. 10 ms^{-1}
4. 20 ms^{-1}

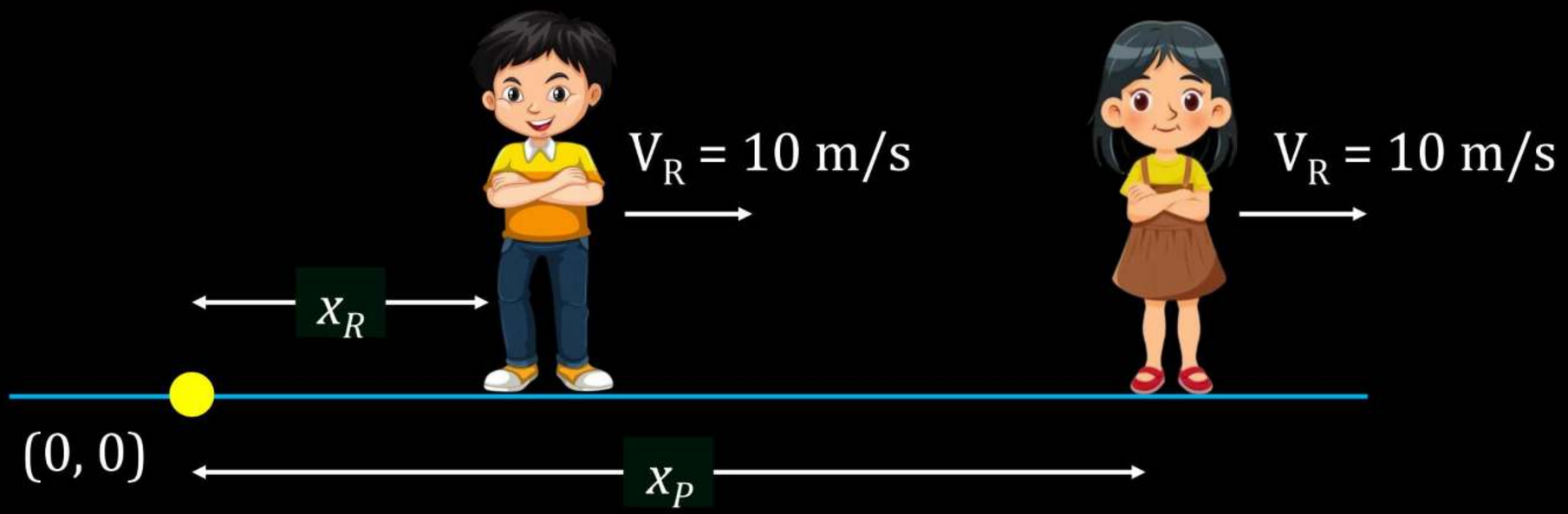
Question

Two cars A and B are moving in same direction with velocities 30 m/s and 20 m/s. When car A is at a distance d behind the car B, the driver of the car A applies brakes producing uniform retardation of 2 m/s^2. There will be no collision when

1. $d < 2.5$ m
2. $d > 125$ m
3. $d > 25$ m
4. $d < 125$ m

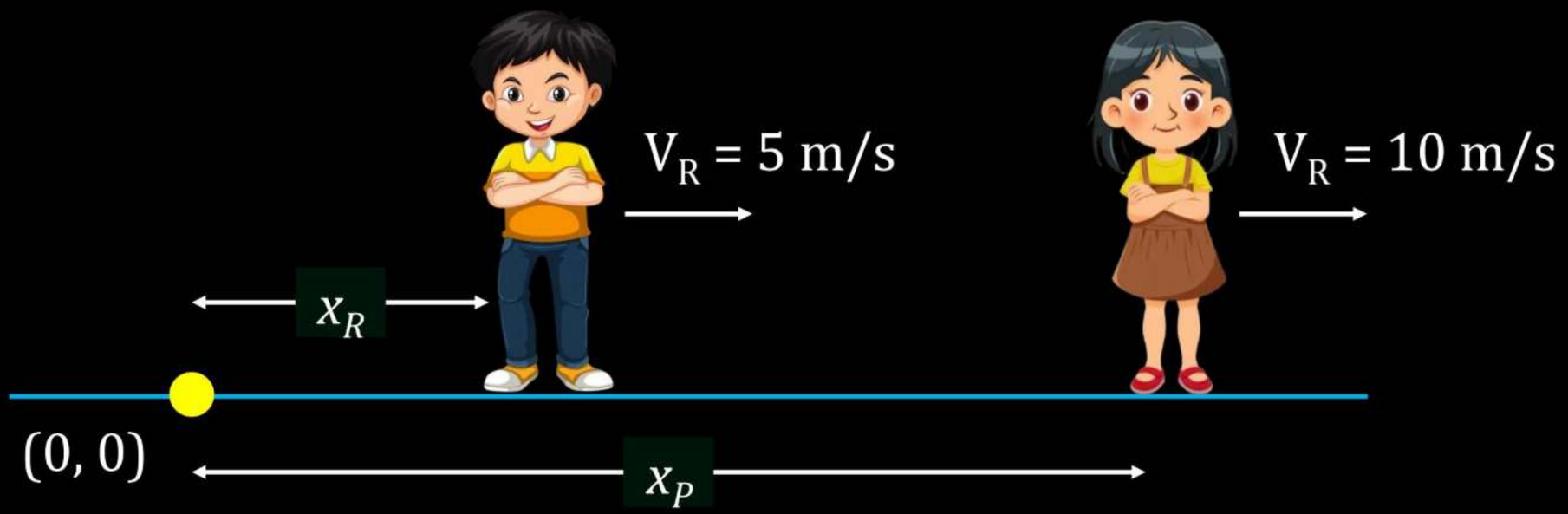
Question

Draw position-time graph and comment on relative separation?



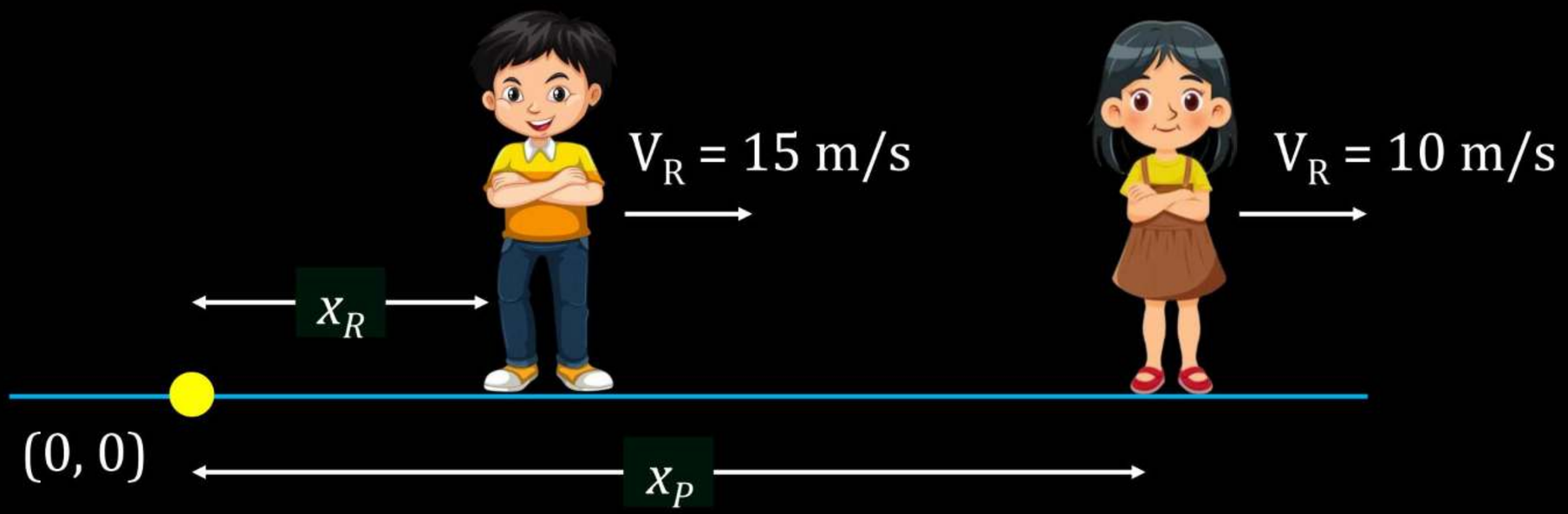
Question

Draw position-time graph and comment on relative separation?



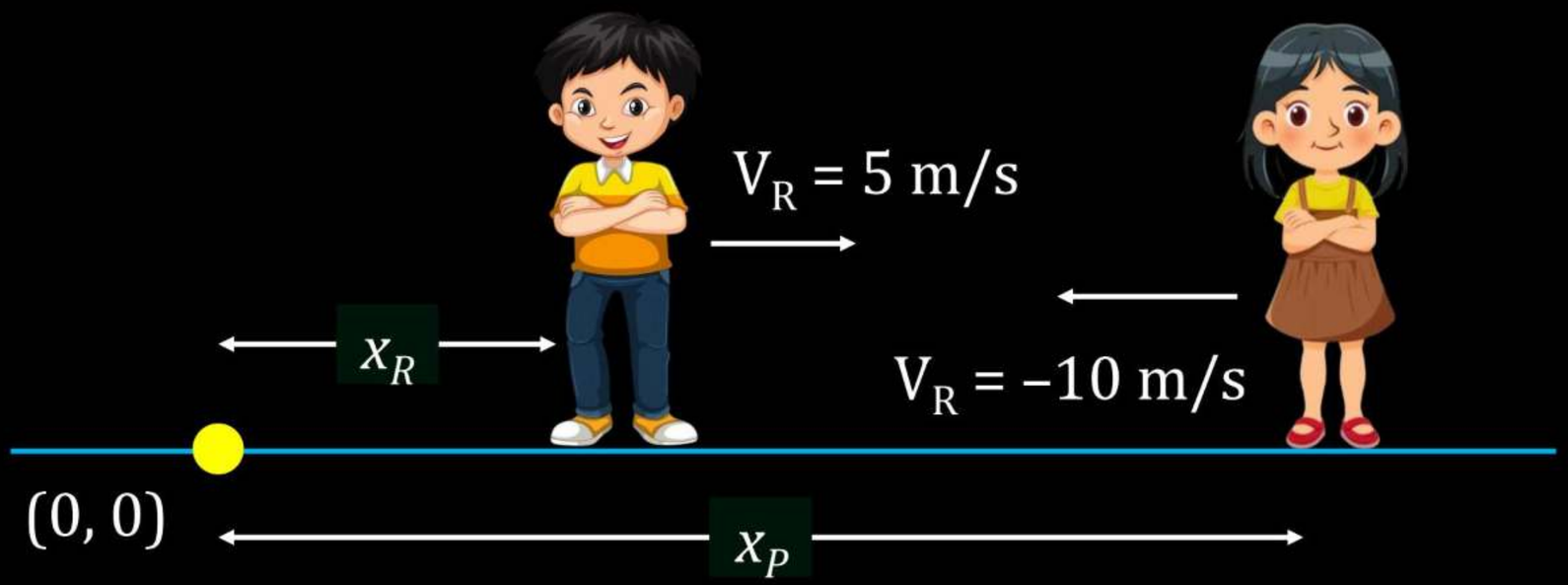
Question

Draw position-time graph and comment on relative separation?



Question

Draw position-time graph and comment on relative separation?



Question

Bus is moving with constant velocity 21 m/s and Ramlal starts his motion from rest and constant acceleration 2 m/s^2. If initial distance is 100 m then find time when Ramlal will catch the bus.

Question

Two cars are moving in the same direction with a speed of 30 km/h. They are separated from each other by 5 km. Third car moving in the opposite direction meets the two cars after an interval of 4 minutes. The speed of the third car is

1. 30 km/h
2. 25 km/h
3. 40 km/h
4. 45 km/h

Question

A ball is dropped from the top of a building of height 80 m. At same instant another ball is thrown upwards with speed 50 m/s from the bottom of the building. The time at which balls will meet is

1. 1.6 s
2. 5 s
3. 8 s
4. 10 s

Question

Find minimum velocity of man so that he can catch the bus who starts motion from rest and acceleration 2 m/s^2.

Question

Two cities X and Y are connected by a regular bus service with a bus leaving in either direction every T min. A girl is driving scooty with a speed of 60 km/h in the direction X to Y notices that a bus goes past her every 30 minutes in the direction of her motion, and every 10 minutes in the opposite direction. Choose the correct option for the period T of the bus service and the speed (assumed constant) of the buses. **[NEET 2025]**

1. 9 min, 40 km/h
2. 25 min, 100 km/h
3. 10 min, 90 km/h
4. 15 min, 120 km/h

Question

Preeti reached the metro station and found that the escalator was not working. She walked up the stationary escalator in time t_1. On other days, if she remains stationary on the moving escalator, then the escalator takes her up in time t_2. The time taken by her to walk up on the moving escalator will be **[NEET -2017]**

1. $\frac{t_1 + t_2}{2}$
2. $\frac{t_1 t_2}{t_2 - t_1}$
3. $\frac{t_1 t_2}{t_2 + t_1}$
4. $t_1 - t_2$



Try all questions ✓

Thank

You